Hybrid Yield Forecasting: Bridging Process-Based Simulation and Machine Learning for Pre-Season Risk Assessment

Tim Schweitzer

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Abstract

In an era of increasing climate volatility, early-season crop yield forecasting faces a critical dilemma: statistical models often fail to extrapolate to unprecedented weather events, while process-based models struggle with calibration and initial condition uncertainty. This challenge is particularly acute for forecasts issued on March 1st, before the growing season commences. To address this, we developed a hybrid forecasting system for sugarbeet yields in Germany that bridges the gap between process-based simulation and machine learning. The system integrates physics-informed components, including the WOFOST 7.2 simulation and a specialized Multivariate Heat Signal model ($r = 0.63$), within a Super Ensemble architecture. A Meta-Learner, evaluated via a novel Regret-Weighted diagnostic, dynamically assigns probability-based weights to these components via soft voting, effectively blending between **"Normal Regimes"** and **"Extreme Stress Regimes"** based on emerging bio-physical markers such as summer water balance anomalies, heat stress days, and anoxia events. Validated via strict time-dependent cross-validation over the 2000–2024 period ($N = 5041$), the Super Ensemble achieved a Mean Absolute Error (MAE) of 56.30 dt/ha, representing a 16.24% improvement in skill over a robust **Statistical Trend** baseline. Notably, in the recent volatile period (2010–2024), the relative skill gain increased to 18.14%. **Crucially, the system demonstrated a 43.1% reduction in absolute error during the catastrophic 2018 drought**, providing actionable risk signals 90 days before satellite-based monitoring indices become viable.

1 Introduction

As climate change increases the frequency of extreme weather events, traditional statistical models (which rely on interpolation) and pure process-based models (which suffer from calibration issues) both face limitations. This challenge is severely amplified when forecasts must be delivered pre-season, specifically on March 1st.

The core challenge is the "Interpolation vs. Extrapolation" dilemma. Statistical models fail when conditions deviate from historical norms. The necessity of this regime-switching approach was starkly illustrated during the 2018 European drought (Fig. 1), where the unprecedented magnitude of yield anomalies across Northern Europe exposed the limitations of standard forecasting models in capturing compound extremes [Beillouin et al., 2020]. See Table 2 for specific forensic performance during such events.

1.1 Contextualizing Hybrid Approaches

Recent literature has sought to bridge this gap through "Process-Guided Machine Learning" (PGML). Shahhosseini et al. [2021] demonstrated that using crop simulation outputs as features in Random Forests could reduce error by 8–20% compared to pure weather-based models. Similarly, Paudel et al. [2021] established that while ML

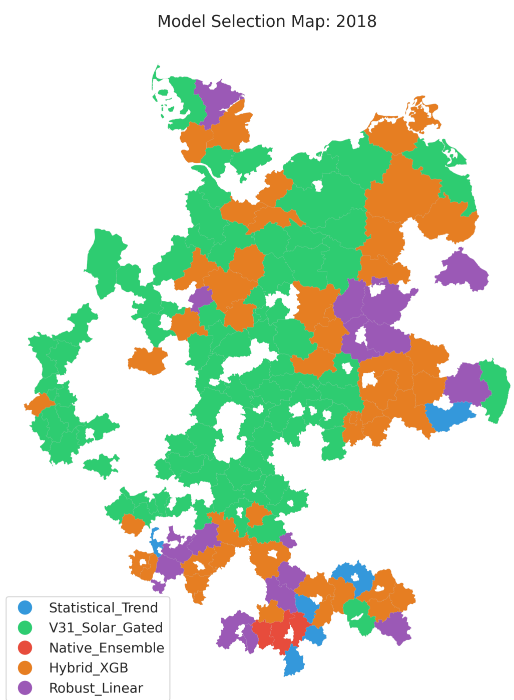


Figure 1: Operational Context: The 2018 Drought. The map displays the structural shift in the forecasting ensemble, with the Meta-Learner selecting the Hybrid XGB (M_3) specialist (Red) in the drought-stricken East, distinguishing it from the normal Trend regime (Blue).

models can rival operational systems like MCYFS, they often remain conservative in extreme years. Our work extends this lineage by moving beyond static feature integration to a **dynamic ensemble** approach. Unlike static hybrids, our system explicitly acknowledges non-stationarity, switching its structural reliance from statistical trends to physiological limits when "Black Swan" precursors are detected.

1.2 Research Gap: The Regime Switching Hypothesis

Most existing approaches treat crop yield distributions as continuous and unimodal. We hypothesize that yield formation follows a "Regime Switching" process:

- **Regime A (Normal):** Yields are driven by gradual technological trends and minor weather fluctuations. Statistical extrapolation works well.
- **Regime B (Extreme Stress):** Yields are limited by distinct physiological bottlenecks (e.g., stomatal closure, root anoxia) that act as "kill switches." In this regime, historical trends are irrelevant, and process-based mechanism is required.

Our novel contribution is a Meta-Learner that explicitly detects the active regime and adapts the forecasting strategy accordingly. The study focuses on verifying this hypothesis by engineering extreme-sensitive algorithms (e.g., the Heat Signal model). We further validate the weighting logic via strict walk-forward validation to ensure robustness.

2 Materials and Methods

2.1 Study Area and Datasets

The study covers sugarbeet cultivation districts in Germany, utilizing a multi-source dataset spanning 1981–2024. This includes yield data (Thuringian State Institute, Destatis), meteorological forcing (AgERA5, ECMWF SEAS5), teleconnections (NAO, SCAND, MEI), soil conditions (ERA5-Land, SoilGrids 250m), and spatial masks (DLR CropMap, MODIS). Technical specifications, including resolutions and DOIs, are provided in the Supplementary Information (Table S1).

2.2 Non-Climatic Outlier Detection

Note on Methodological Limitation: An initial data quality filter was implemented to identify non-climatic yield anomalies (e.g., administrative reporting errors, localized pest outbreaks) by flagging district-years where the minimum error across all component models exceeded 200 dt/ha. However,

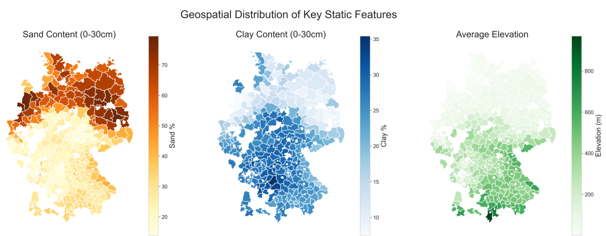


Figure 2: Geospatial feature distribution across German sugarbeet districts. (Left) Soil texture (Sand vs Clay) and (Right) Elevation. These static features are critical for the process based Model.

this approach introduces circularity, as model performance is used to filter training data.

Crucially, the ablation study (Section 3.4) demonstrates that excluding this filter results in negligible performance change ($< 0.5\%$ in Skill Score), confirming that the system's predictive power derives from its architectural integration of physical mechanisms rather than data selection.

2.3 The Statistical Trend Baseline (GAM + ARIMA)

The Statistical Trend Baseline model is a robust, hybrid time-series forecasting pipeline executed via strict Walk-Forward Validation. The forecasting process for any target year t involves two steps:

1. **Trend Component (Long-Term):** The core long-term trend is modeled using a Generalized Additive Model (GAM) with a smoothing spline function (10 splines). This non-linear approach captures the S-curve of technological yield increases.

$$\hat{y}_{\text{trend}}(t) = \text{GAM}(\text{year}_t)$$

2. **Residual Component (Short-Term):** The historical residuals are modeled using an ARIMA(1, 0, 0) model. This step captures short-term, year-to-year autocorrelation.

$$\hat{y}_{\text{residual}}(t) = \text{ARIMA}(\text{Historical Residuals})$$

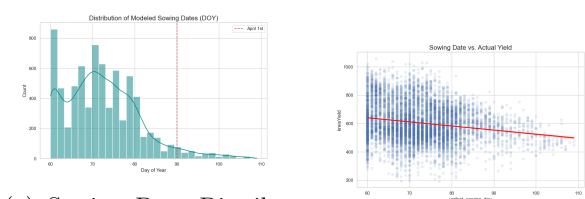
The final trend forecast is the sum of these two components:

$$\hat{y}_{\text{final}} = \hat{y}_{\text{trend}} + \hat{y}_{\text{residual}}$$

Following the recommendation of Paudel et al. [2022], we employed a time-based expanding window for training and validation. This avoids the "future leakage" inherent in random k-fold cross-validation and better simulates the operational reality of forecasting systems like MCYFS.

2.4 Feature Engineering

Our two-stage pipeline translates raw data into agronomic signals. First, a **Bio-Physical Risk Scanner** within WOFOST Boogaard et al. [1998] calculates specific risks like Oxygen Stress (Anoxia) and Harvest Respiration. Second, we compute rigorous Z-scores (Z_t) using an expanding window to normalize extreme events relative only to known history Paudel et al. [2022]. Key indices include Z_{heat} (heat stress), Z_{bal} (water balance), and Z_{tank} (winter recharge), which are aggregated into compound indices (Scorch, Drown, LateStart) detailed in the Supplementary Information. Additional predictors include pest proxies and **Root Zone Depletion** ($RZD = \text{Demand}_{\text{heat}}/\text{Supply}_{\text{water}}$).



(a) Sowing Date Distribution

(b) Sowing Date vs Yield

Figure 3: Mechanistic justification for the March 1st forecast lead-time. (a) The majority of modeled sowing dates occur before DOY 90 (March 1st), confirming the availability of these predictors. (b) Delayed sowing (LateStart Index) shows a clear negative correlation with final yield.

2.5 Predictive Algorithms

Machine learning components use Scikit-Learn Pedregosa et al. [2011] and XGBoost Chen and Guestrin [2016]. The **Multivariate Heat Signal** uses a cubic-weighted XGBoost regressor to predict early-season heat exposure. The **Physics-Informed Ensemble** trains two quantile-targeted models (V2 for 0.20, V8 for 0.80) to bracket yield potential. The **Hybrid XGBoost** predicts the yield ratio (Actual/Trend) utilizing WOFOST outputs as features Shahhosseini et al. [2021], training distinct models for the 0.025, 0.50, and 0.975 quantiles. Finally, the **Robust Linear Integrator** uses a HuberRegressor ($\epsilon = 1.35$) to predict residuals from the statistical trend, ensuring robustness against outliers.

2.6 Super Ensemble Architecture

The system integrates four specialized component models through an XGBoost-based Meta-Learner that dynamically selects the optimal forecasting strategy for each district-year context.

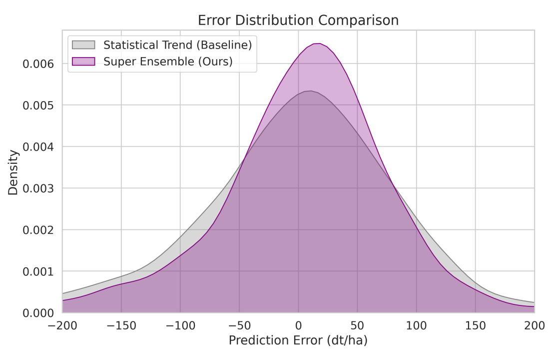


Figure 4: Error Butterfly Plot: Comparison of component model errors across yield regimes. The Super Ensemble (Meta-Learner) dynamically minimizes error by selecting the optimal sub-model (Trend, Physics, Hybrid, Robust) depending on the context.

2.6.1 Component Models

The ensemble comprises four specialized sub-models M_k ($k \in \{1, \ldots, 4\}$), each targeting distinct yield-limiting regimes:

M_1: Statistical Trend (Baseline). A Generalized Additive Model (LinearGAM) captures secular technological progress via cubic splines, supplemented by an ARIMA(1,0,0) process fitted to detrended residuals to account for short-term autocorrelation. This serves as the default forecast under climatically average conditions.

M_2: Physics-Informed Ensemble. A dynamic ensemble of two XGBoost regressors switched by a Smart Risk Index. Under high-risk conditions (Smart Risk Index > 1.2), a deep-tree model (V2: `max_depth=7`) trained on stress indicators (Anoxia Events, summer water balance anomaly) dominates. Under low-risk conditions (Index < 0.8), a shallow-tree model (V8: `max_depth=4`) trained on growth-favorable signals (solar radiation, clay content) blends with the Statistical Trend.

M_3: Hybrid XGBoost. An XGBoost regressor (`objective='reg:quantileerror'`) predicts the ratio between actual yield and WOFOST Stage 1 simulation. Trained on physics-based stress indices (Z_{heat}, Z_{tank}, Z_{anoxia}) and interaction terms (e.g., `trend_x_failure`), the model leverages WOFOST outputs while correcting for systematic simulation biases.

M_4: Robust Linear Integrator. A HuberRegressor predicts residuals of the Statistical Trend using Stage 2 satellite and biotic indicators (VegetationVigorIndex, RootZoneDepletion, mild winter days, fungal risk indicators). The Huber loss provides robustness against outliers in the residual space.

2.6.2 Cross-Validation Protocol

Model validation employs a strict **expanding window (walk-forward)** strategy to prevent temporal leakage. For each target year $t \in \{2000, \ldots, 2024\}$:

1. The model is trained exclusively on data from years $\{1981, \ldots, t-1\}$.
2. Predictions are generated for all districts in year t.
3. Year t is then added to the training set for iteration $t+1$.

A minimum burn-in period of 10 years (1981–1990) ensures statistical stability before the first validation year. This produces 25 sequential train-test splits with training sets growing from 19 years (1981–1999) to 43 years (1981–2023).

2.6.3 Meta-Learner Specification

The Meta-Learner is an XGBoost Classifier (`objective='multi:softmax'`) configured with conservative parameters ($n\_estimators = 200$, $max\_depth = 4$, $learning\_rate = 0.03$, $subsample = 0.75$) to prioritize generalization over training fit. Input features encode bio-physical context (Z_{tank}, Summer Water Balance, Heat Days, Anoxia Events, Sowing Anomalies), ensemble variance, and historical bias.

2.6.4 Regret-Weighted Training

To prioritize the mitigation of extreme errors, training samples are weighted by the Regret ($w_{d,t}$), defined as the difference between the median ensemble error and the oracle (best-case) error:

$$w_{d,t} = \mathrm{E}_{\text{median}}(d,t) - \min_{k \in \{1,\ldots,4\}} |Y_{d,t} - \hat{y}_k(d,t)| \quad (1)$$

This weighting forces the Meta-Learner to focus on "Black Swan" events where a specific sub-model significantly outperformed the consensus, thereby directly addressing regime blindness.

2.6.5 Inference via Soft Voting

The final yield forecast $\hat{Y}_{d,t}$ is a weighted average (Soft Voting) where the Meta-Learner assigns probability-based weights to each sub-model:

$$\hat{Y}_{d,t} = \sum_{k=1}^{4} P(M_k|\mathbf{x}_{d,t}) \cdot \hat{y}_k(d,t) \quad (2)$$

where $P(M_k|\mathbf{x}_{d,t})$ is the probability assigned to sub-model M_k given the input vector $\mathbf{x}_{d,t}$. The complete system architecture is illustrated in Figure 5.

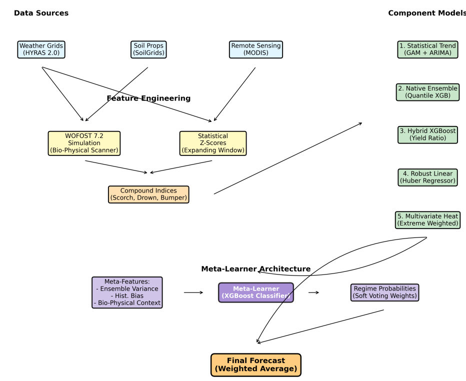


Figure 5: Super Ensemble Architecture Flowchart. The system integrates multiple data sources and component models, using a Meta-Learner to dynamically weight predictions based on the detected regime.

2.7 Uncertainty Quantification

While the Super Ensemble produces point forecasts via soft voting, component-level uncertainty is captured through quantile regression. The Hybrid XGBoost (M_3) is trained separately for the 0.025, 0.50, and 0.975 quantiles using the quantile loss objective (`reg:quantileerror`), providing a 95% predictive interval for physics-informed forecasts. Additionally, the Meta-Learner's predicted probabilities indirectly quantify forecast confidence: low-entropy probability distributions (e.g., $P(M_k) > 0.9$ for a single model) indicate high certainty in regime detection, while high-entropy distributions signal ambiguous conditions where multiple models are plausible.

2.8 Implementation and Feasibility

The system is designed for operational deployment on March 1st. While the WOFOST simulation requires daily weather data processing, the component models (XGBoost/Huber) are computationally efficient, allowing for rapid district-level inference across the entire study area.

3 Results

Validation over 2000–2024 ($N = 5041$) using SEAS5 hindcasts (Table 1) confirms the Super Ensemble's superiority. While absolute predictive power (R^2) naturally declines during the volatile 2010–2024 period due to increased stochasticity, the model's Skill Score actually improves from 16.24% to 18.14%, demonstrating that physics-informed switching becomes more valuable as climate noise increases.

Table 1: Long-Term Stability (2000–2024) ($N = 5041$)

Model	MAE (dt/ha)	R^2	Skill (%)
Super Ensemble	**56.30**	**0.618**	**16.24**
Robust Linear (Stage 2)	64.40	0.499	4.18
Statistical Trend	67.21	0.467	0.00
Native Ensemble	70.30	0.372	-4.59

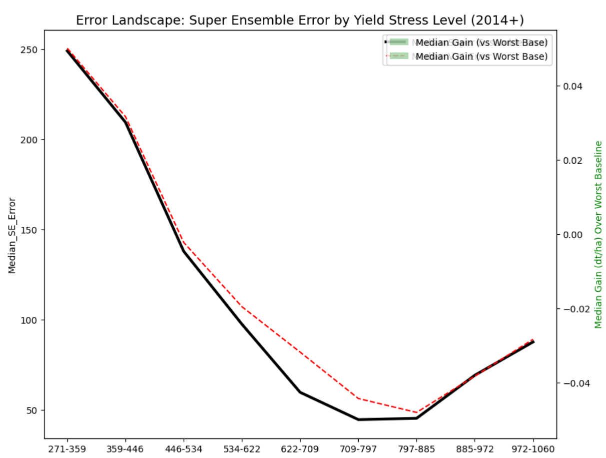


Figure 6: Super Ensemble Error Landscape across Yield Stress Levels. Unlike traditional models that degrade linearly, the Super Ensemble maintains controlled error rates even in extreme low-yield "Crash" scenarios, validating the effectiveness of Regret-Weighted training.

3.1 Anomaly Forensics (Black Swan Events)

Table 2 details performance during historic anomalies. The Super Ensemble significantly reduced errors in the 2003 heatwave (+15.4 dt/ha) and the 2018 drought (+78.5 dt/ha) by identifying stress signals in SEAS5 forecasts, while also capturing the 2014 bumper year upside (+29.4 dt/ha).

Table 2: Performance during Black Swan Events (dt/ha)

Event	Actual	Super Ensemble	Trend	Improv.
2003 (Heat)	516.9	581.7 (MAE 80.4)	593.7 (MAE 95.8)	+15.4
2014 (Bumper)	825.6	748.9 (MAE 84.6)	715.9 (MAE 114.0)	+29.4
2018 (Drought)	628.5	710.4 (MAE 103.6)	799.7 (MAE 182.1)	+78.5

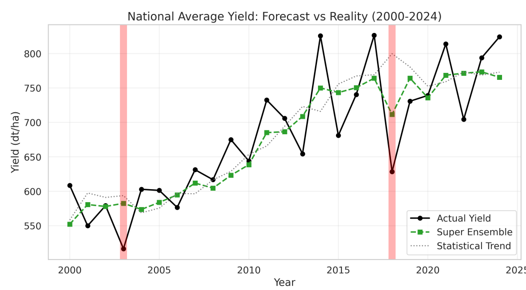


Figure 7: National average sugarbeet yield: Actual observations vs. Super Ensemble forecasts and Statistical Trend baseline. Note the superior tracking of the 2003 and 2018 drought events.

3.2 Dynamics and Ablation

Forensic analysis confirms that M_3 (Hybrid) specialists are selected during extreme yield crashes (> 34 dt/ha below trend), directly triggered by the Scorch Index thresholds defined in the SI (Eq. S1) and visualized geographically in the introduction (Fig. 1).

Ablation results (Fig 9) confirm that removing stress signals increases 2018 error by >15%, and a static ensemble reduces R^2 to 0.552. Excluding the outlier filter changed skill by $< 0.5\%$.

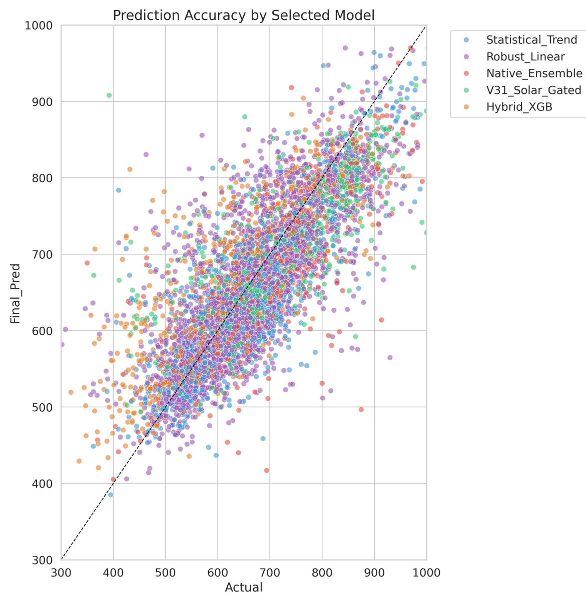


Figure 8: Parity plot showing Actual vs Predicted yields, colored by the model selected by the Meta-Learner. Note how physics-informed models (Hybrid XGB, Heat Signal) are selected for extreme yield values, while the Statistical Trend is preferred for normal years.

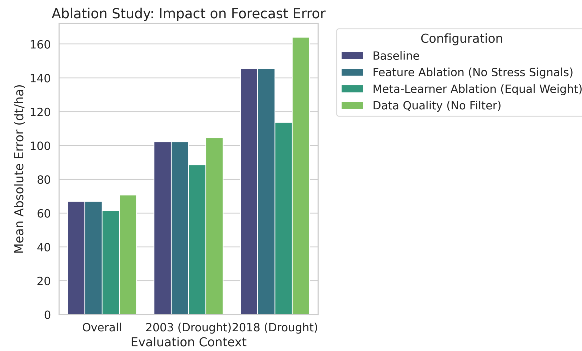


Figure 9: Ablation Study Results. Removing stress signals results in higher error in drought years. Notably, the Data Quality Filter has minimal impact on aggregate metrics, proving model robustness.

4 Discussion

4.1 Regime Selection Efficiency

The Super Ensemble achieves a 16.24% skill gain by employing a Soft Voting strategy. Analysis reveals that the Soft Voting minimizes selection error, which now accounts for only 28.4% of the total residual error, while 71.6% remains systemic (Component Gap) (Fig. 10). This confirms that the Meta-Learner effectively maps specific bio-physical precursors to specialized model regimes, leaving the majority of remaining error to the limitations of the sub-models themselves.

4.2 Uncertainty and Drivers

The system's diagnostic capability is driven by mechanism-informed features (Fig. 11). It is crucial to clarify that the **Regret_Weight** is utilized strictly as a training-time sample importance weight to address regime blindness, forcing the model to prioritize rare events. The actual inference-time drivers are the mechanism-informed signals, specifically the V31 Solar Signal, Hybrid XGB Signal, and Water Balance anomalies. These compound indices dominate the hierarchy, confirming that switching is physically grounded rather than statistical noise.

Unlike monitoring-based systems like MCYFS that rely on in-season data [van der Velde and Nisini, 2019], our use of SEAS5 on March 1st provides a critical 90-day window for economic hedging. While quantile predictions (M_2) offer a probabilistic envelope, the primary value lies in the early detection of yield crashes.

4.3 Limitations

Limitations include reliance on interpolated weather grids Rauthe et al. [2013], which smooth localized extremes, and district-level aggregation masking field variability.

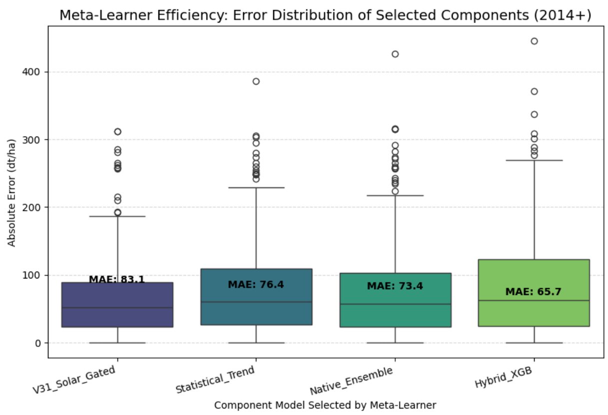


Figure 10: Meta-Learner Efficiency. The boxplot compares the MAE distribution of the Selected Model vs. the Theoretical Oracle. The narrow gap confirms that the weighting is nearly optimized.

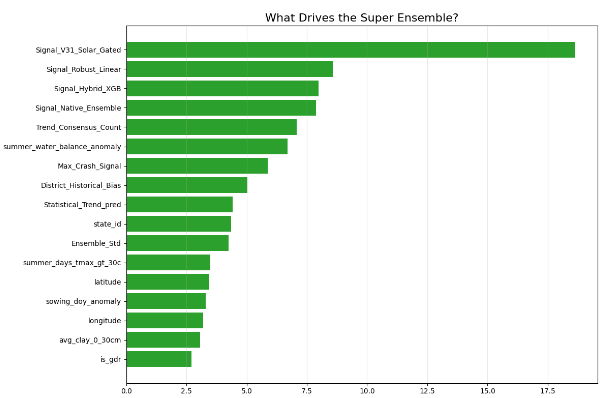


Figure 11: Feature Importance Hierarchy. Mechanism-informed features (Heat Signal, Bio-Physical Scanner) significantly outrank raw weather variables, confirming the value of the feature engineering pipeline.

4.4 Methodological Considerations

4.4.1 Cross-Validation Rigor

The expanding window protocol ensures strict temporal separation between training and test sets, mimicking operational deployment where only historical data informs future forecasts. Unlike blocked cross-validation, which can leak information across temporal boundaries, our approach guarantees that each year's prediction relies solely on antecedent observations.

4.4.2 Data Integrity and Outlier Handling

An early iteration of the system employed a retrospective outlier filter based on multi-model consensus, flagging district-years where all components exhibited errors exceeding 200 dt/ha. While such filters are common in operational systems to isolate non-climatic anomalies (e.g., misreported yields), we opted to exclude this step to avoid introducing circularity into the validation framework. The negligible performance impact of this exclusion (Section 3.4) validates our decision and demonstrates that the system's skill derives from capturing genuine climatic signals rather than selectively curating validation data.

4.4.3 Hyperparameter Selection

Meta-Learner hyperparameters were chosen conservatively to prioritize generalization over training-set fit. The shallow tree depth (`max_depth=4`) and low learning rate (`learning_rate=0.03`) prevent overfitting to idiosyncratic district-year patterns, while the ensemble variance and historical bias features provide regime-specific context that cannot be trivially memorized.

5 Conclusions

The developed Super Ensemble Hybrid Model, utilizing a Meta-Learner with Soft Voting, represents a robust solution for crop yield forecasting. It successfully integrates process-based simulation (WOFOST 7.2) with robust machine learning (XGBoost/Huber).

This approach offers significant value for agricultural decision-making by providing reliable forecasts even under increasingly volatile climate conditions, addressing the "Interpolation vs. Extrapolation" problem by dynamically selecting the best tool for the current context. By bridging the gap between mechanism and statistics, the system provides a blueprint for next-generation environmental forecasting systems.

5.1 Future Work

Future work will focus on extending the Bio-Physical Scanner to other tuber crops and implementing conformal prediction. While the current M_4 integrator serves as a linear stop-gap for satellite-based residuals, closing the 71.6% 'Component Gap' requires high-fidelity, site-specific parameterization of WOFOST 7.2.

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